Lesson 5
Introduction to Simulation

E211 – Operations Planning II

Learning Outcomes

- Interpret the meaning of ‘System’ in simulation context.
- Identify when to use simulation to study a problem.
- Describe the steps in carrying out a simulation study.
- Describe the pitfalls, advantages and disadvantages of simulation.

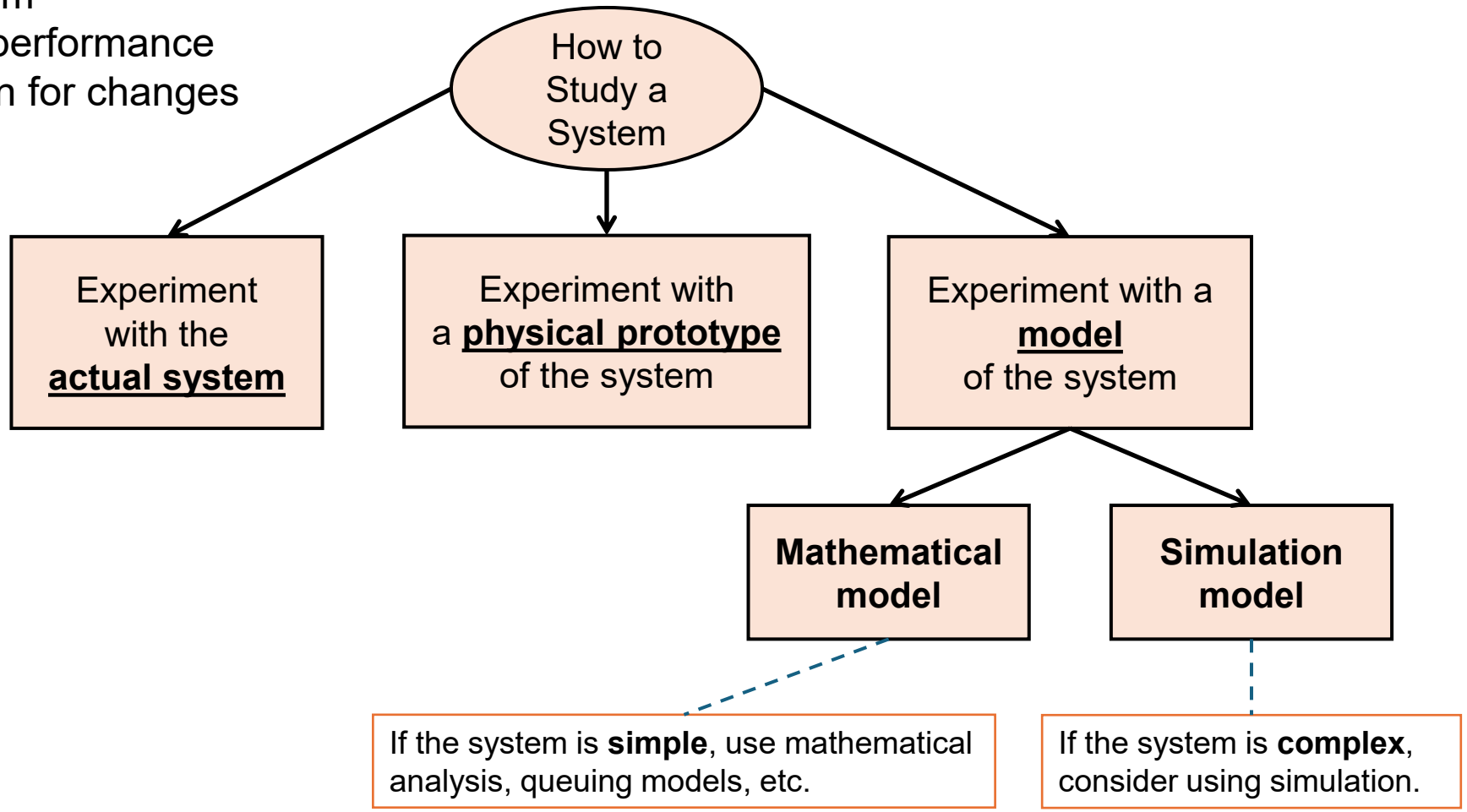
What is a System?

- A system is a facility or process with interacting components operating together to achieve a common goal or objective.
- Examples of real-world systems:
  - A **manufacturing facility** with machines, conveyor belts, work-in-process parts, production schedule and inventories.
  - A **distribution centre** with customer service, warehousing, shipping and delivery teams.
  - A **hospital** with wards, A&E department, clinics, pharmacies, linen/ logistics department and kitchen.

ALL REAL-WORLD SYSTEMS CONTAIN QUEUING PROCESSES AND MANY ARE HIGHLY COMPLEX!

When to use Simulation to study System

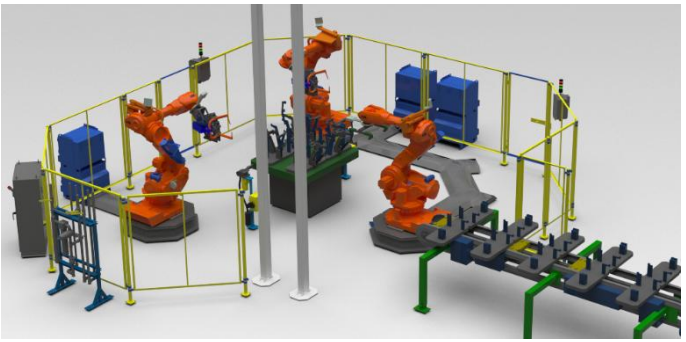
Purpose of studying a system
- Measure/ predict system performance
- Improve operations or plan for changes



What is Process Simulation?

- Use of **computer models** to mimic **real-world** processes or systems over time
  - Powerful tool for studying complex systems
  - Require use of computer software to model and simulate
  - Scientific, data-based tool for decision making
  - Predict answers to ‘what if’ questions

- Real world applications such as:
  - Design of manufacturing/ distribution/ healthcare processes
  - Planning of facility layout
  - Analysis of supply chain operations
  - Improvement study of airport operations



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Types of Simulations

Static (Monte Carlo)

Simulation predicts system at a particular point in time. Doesn't track changes over time.

E.g.

Project management risk analysis

Financial investment forecasting

Dynamic (track changes of system over time)

Continuous Simulation (System properties change continuously with time):

- Differential Equations

  Eg. Chemical reactions, weather patterns

- Kinematics

  E.g. Robot arm movement, rocket trajectory

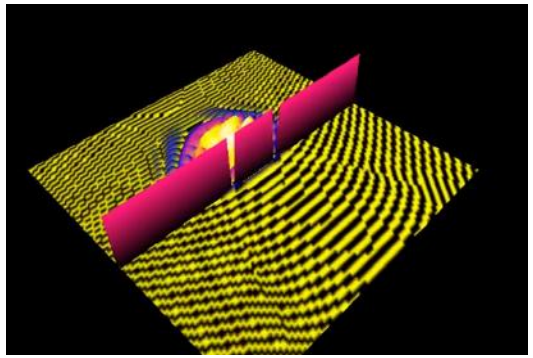
Discrete Event Simulation (System properties change at specific points in time):

- Used to depict queuing systems in industrial operations,
- E.g. Process simulation of manufacturing lines, warehouse operations

Identify the Type of Simulation (SCORM Quiz)

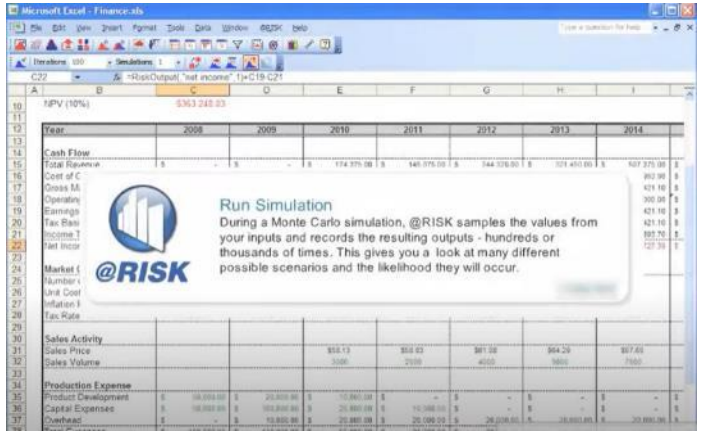
Video 1

Video 2



Monte Carlo, Continuous or Process simulation?

Video 3



Video 4

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Components of a Discrete Event Simulation (I)

Entity: An object of interest in the system

- *Dynamic objects* - get created, move around, change status, affect and are affected by other entities, leave (maybe)
- Can have different types of entities concurrently
  - E.g. Patients, Medical supplies

Attribute: A characteristic of all entities, but with a specific value "local" to the entity that can differ from one entity to another.

- E.g. Patient's age, illness

Resources: What entities compete for

- Entity *seizes* a resource, uses it, releases it
- A resource can have several units of capacity which can be changed during the simulation
  - E.g. Doctors, nurses, X-ray machines

Components of a Discrete Event Simulation (II)

Event: An instantaneous occurrence that changes the state of the system

- E.g. Arrival of a patient, end of treatment, failure of a machine

Activity: Represents a time period of specified length or a collection of operations that may transform the state of an entity

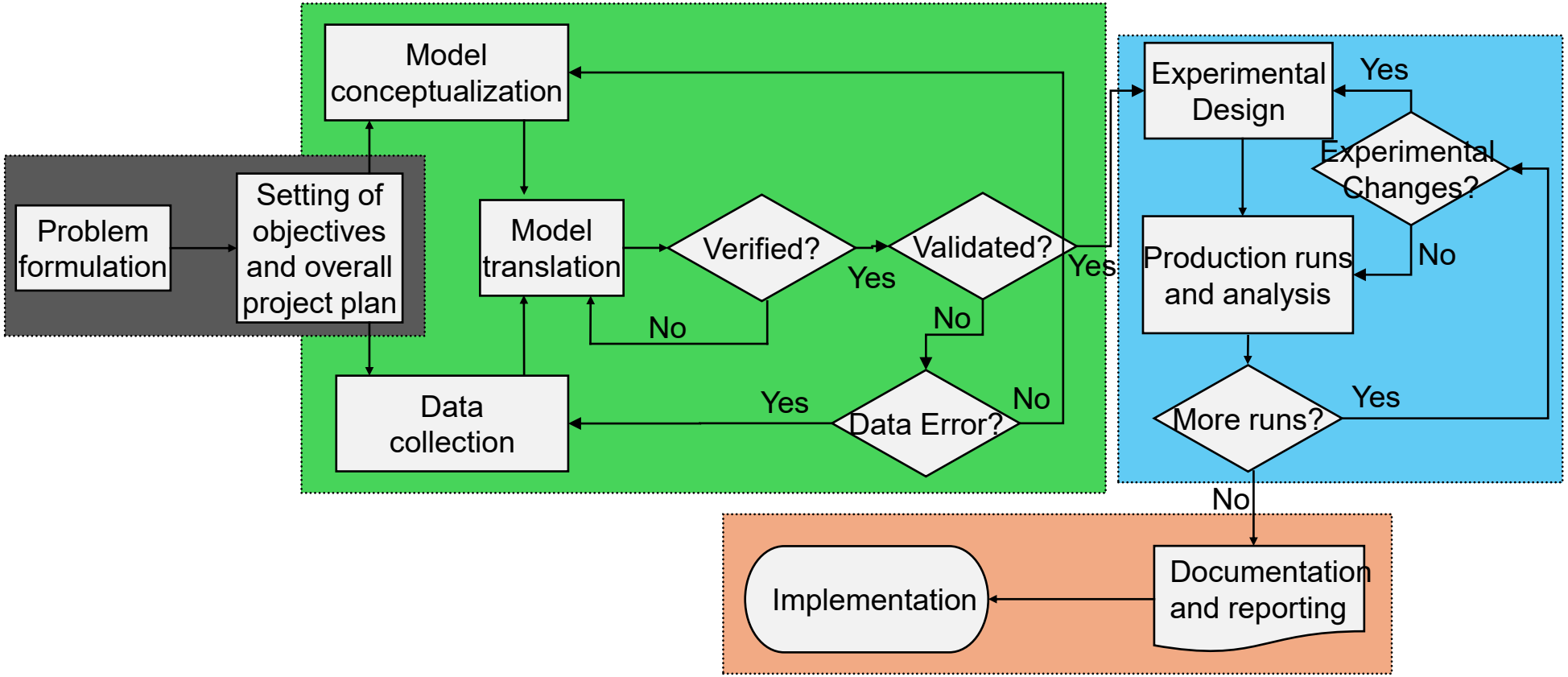
- E.g. X-ray scan, doctor consultation, making payment, etc.

State: A collection of variables that contains all the information necessary to describe the resources/entities/system at any time

- E.g. Current state of patient (waiting, under treatment, walking, etc)
- E.g. State of doctor (busy, idle, rest, etc)

Identify Components of a Discrete Event Simulation (SCORM quiz)

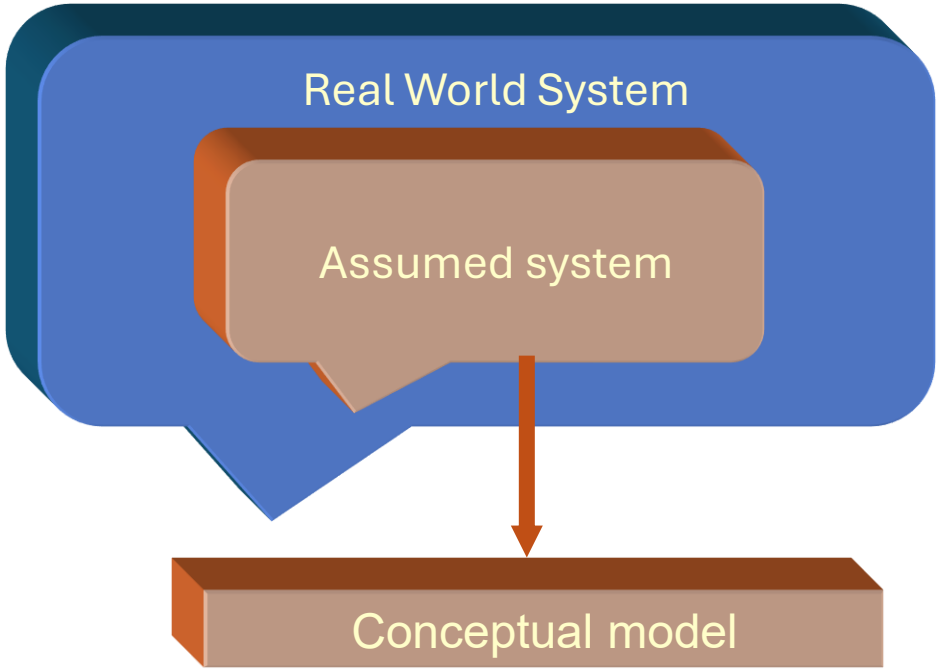
Steps of a Simulation Study



Step 1: Problem Formulation

- Kickoff meeting with project manager, simulation analysts and subject-matter experts (SME) to define:
  - Objectives: questions to be answered by the study
  - Performance measures to be used to evaluate the system performance
  - Scope of model and level of details
  - Time frame for the study and required resources

Step 2: Model Conceptualization



- Conceptual model is the documented representation of the system where the boundary and assumptions of the model are defined
- Often use process maps or flowcharts to show flow of the entities through various activities and the key events
- The output and performance measures of the system are defined
- The data to gather such as process timings and resource deployment are defined

Level of Detail in Model

- Scope the study at the appropriate level of detail

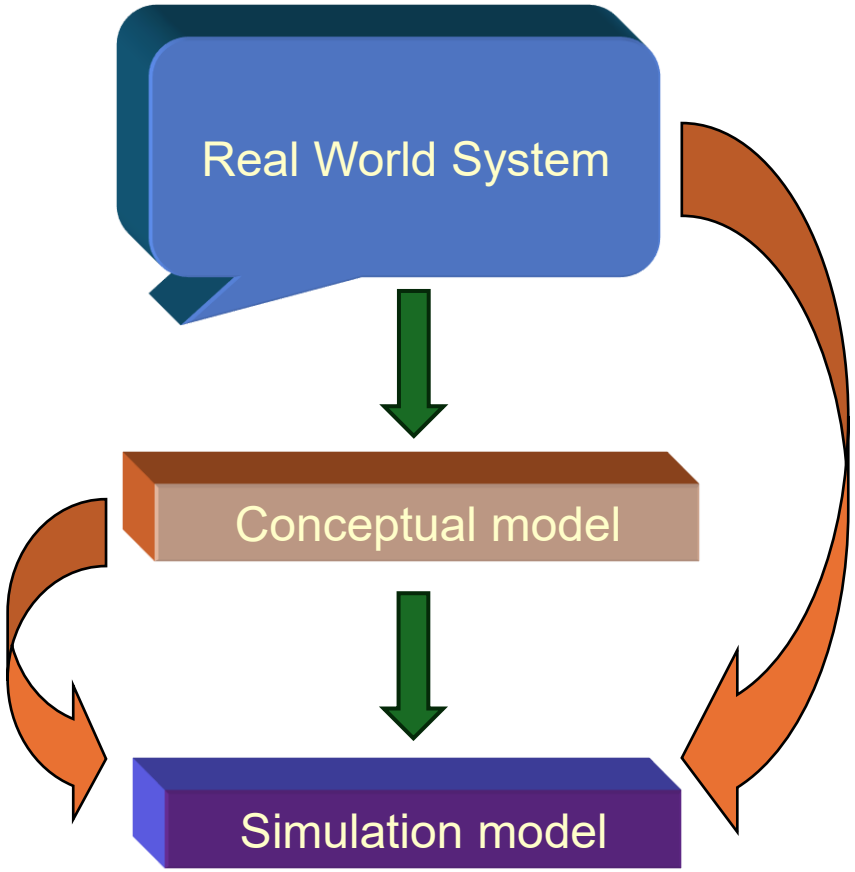
- Model contains too much information
  - Greater time and effort
  - Longer simulation runs
  - More likely to contain errors

- Model contains too little information
  - Key info may be missing
  - Result or recommendation may not be valid

Step 3: Model Translation

- Build the model using a simulation software
- State-of-the-art simulation software include:
  - **FlexSim**
  - Arena
  - Witness
  - Simio
  - Anylogic

Step 4: Verification and Validation



Validation:

The process of determining if the model accurately represents the system.

I.e. Comparison of model results with real-world system performance

Verification:

The process of determining if the simulation logic is correct.

I.e. Debugging the simulation model

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Step 5: Experimental Design & Analysis

- Experimental design and modeling
  - ❑ Are there alternative system configurations that may improve the current situation?
  - ❑ Could changing the process flow improve the problem?
  - ❑ ‘What if’ scenarios – how will changes to input parameters such as arrival rates, resource levels and process timings affect performance?
- Conduct simulation runs and analyze output
  - ❑ Compare results from each scenario

Step 6: Documentation and Reporting

- Program Documentation
  - Document assumptions and study's results to facilitate recommendations
  - Discuss model building and validation process to promote credibility
  - Use animation to communicate to managers and others who are not familiar with the system
  - Results will be used in decision making if they are valid and credible.

Pitfalls to Successful Simulation

- Failure to have a well-defined set of objectives at start of study
- Inappropriate level of model detail or insufficient information and data to define the scope
- Failure to communicate with management throughout course of study
- Treating a simulation study like an exercise in computer programming
- Failure to uncover key information and validate the model
- Failure to have people with knowledge in simulation methodology and statistics

Advantages of Simulation

- ✓ When mathematical analysis methods are not available, simulation may be the only investigation tool
- ✓ Allows comparisons of alternative designs or operating policies before physical changes to the system
- ✓ Allows users to visualize process flows and system changes clearly in a 3D environment
- ✓ A convincing risk management and decision-making tool through study of ‘what if’ scenarios

Disadvantages of Simulation

- Simulation forecasts the output of a model. It does not generate or suggest a solution.
- Simulation of a complex system may be expensive and time consuming to develop. Often requires a multi-disciplinary team to carry out a proper simulation study.
- An invalid model may result in having confidence in wrong results. Validity of model depends on quality of input data.

FlexSim Introduction Video

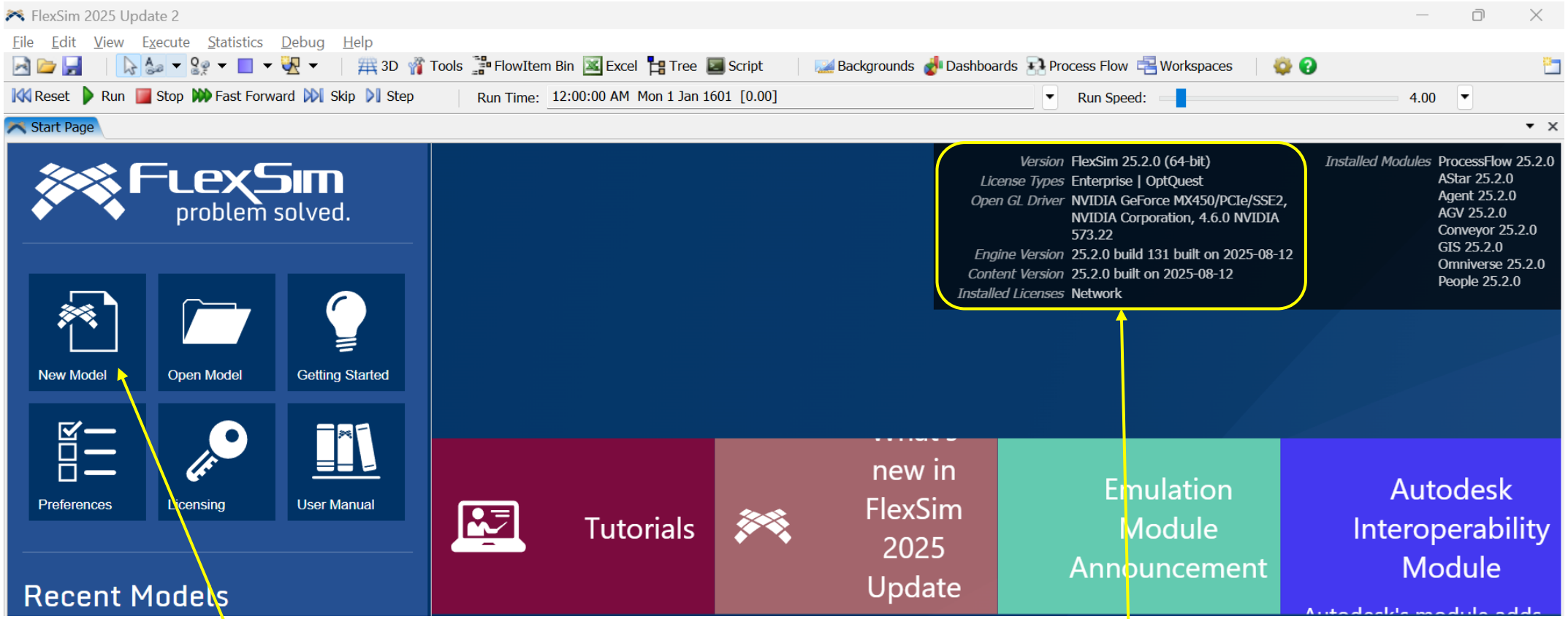
Refer to ‘AY 2025 Sem 2 E211 FlexSim 25.2.0 Installation’ pdf on how to install FlexSim.

More info @ www.flexsim.com

FlexSim Hands-On:

Build Your 1st FlexSim Model

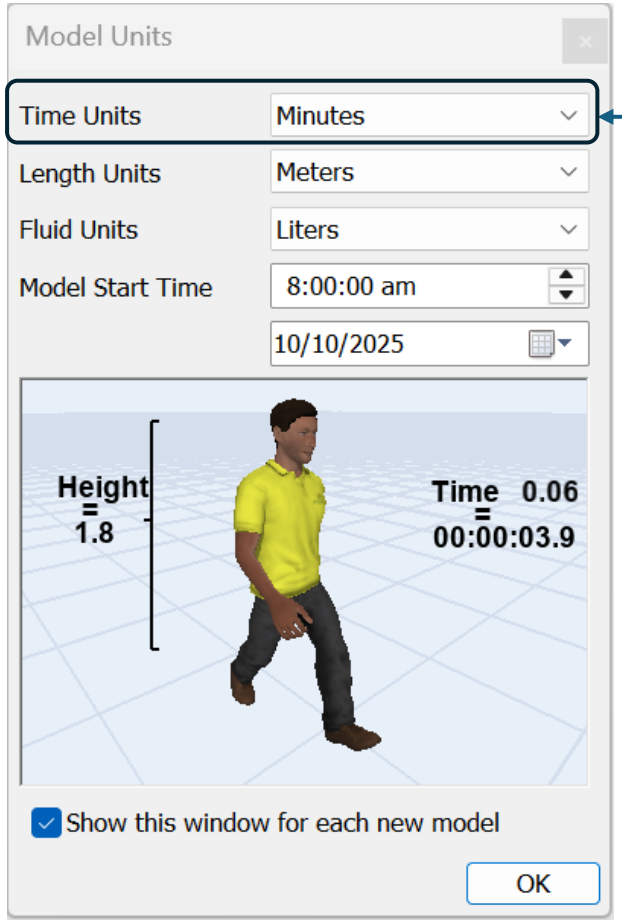
FlexSim Start Page



Click New Model

Check the version and license

Model Units Window



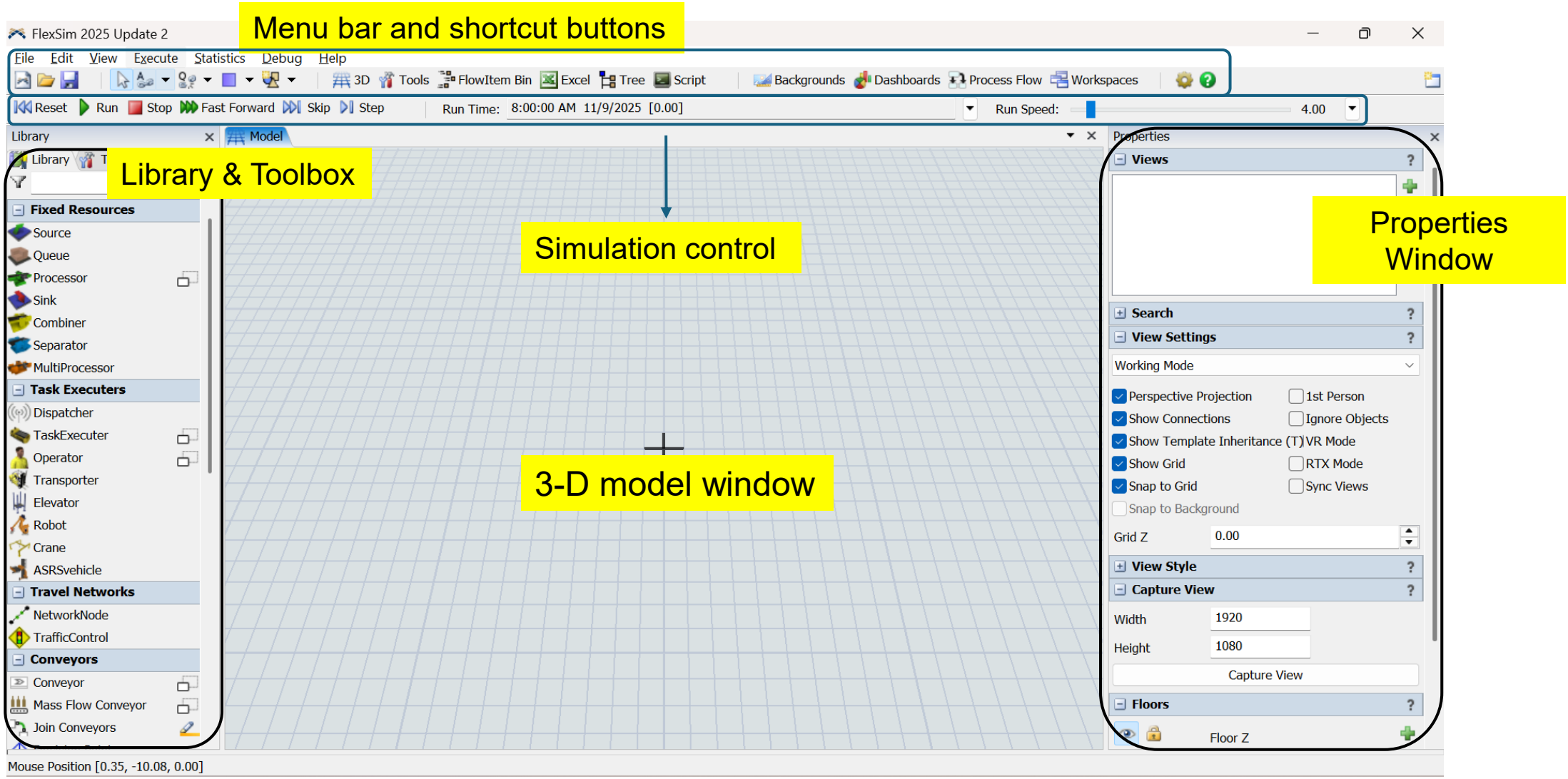
Set time unit to 'minutes'

When setting model units, consider:

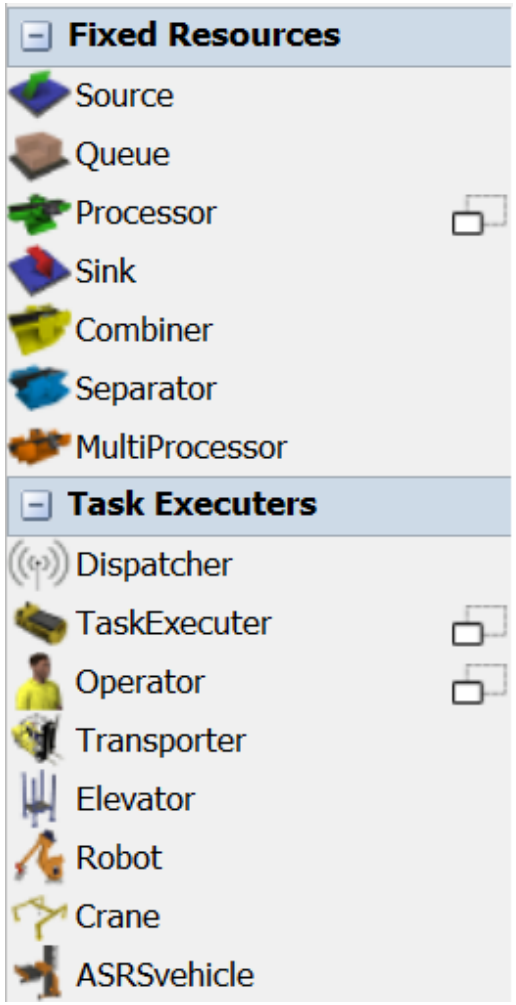
Time Units – should the simulation time run in seconds, minutes or hours?

Length Units – what is distance unit you want to one grid in the model to represent?

FlexSim Environment



FlexSim 3D Object Library



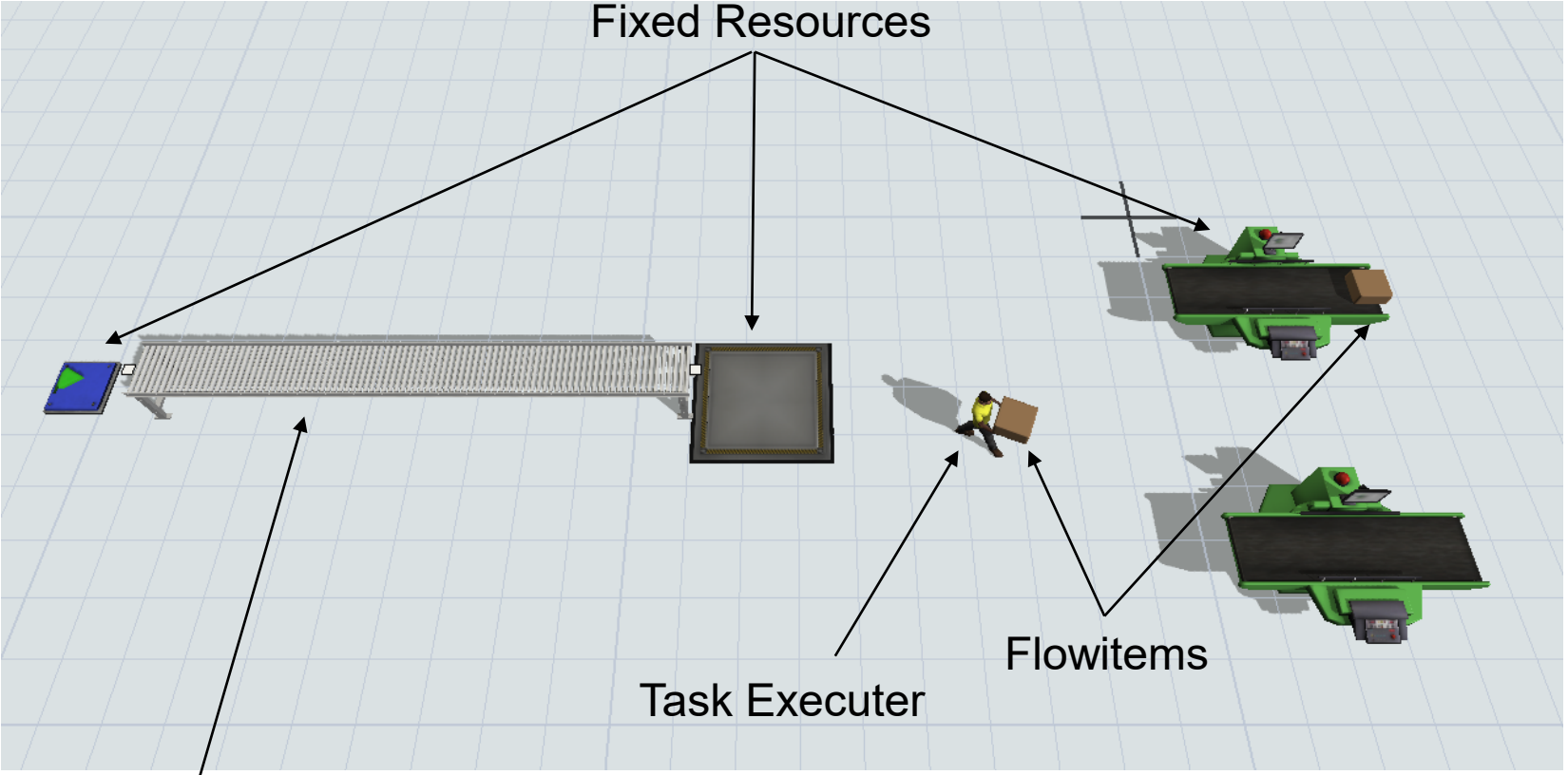
Fixed Resource:

- Stationary. Transforms flowitems
  - Generates items
  - Holds items
  - Processes items
  - Destroys items

Task Executer:

- Can move
  - Transports flowitems
  - Performs tasks

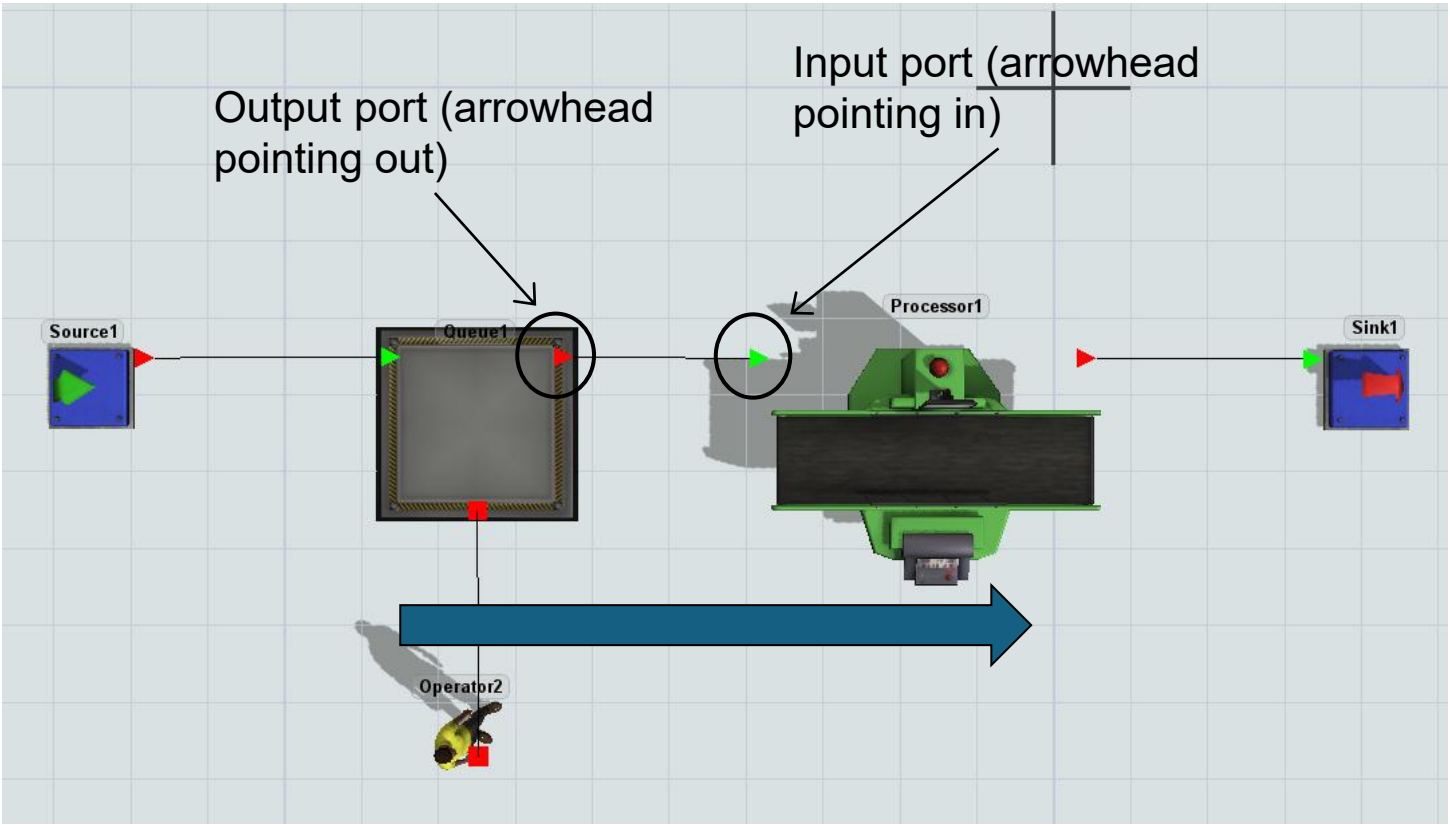
Objects and Entities in FlexSim



Flowitems:

- Entities that flow through the process
- Can represent physical objects or people
- Generated in Source and destroyed in Sink

Direction of Flow



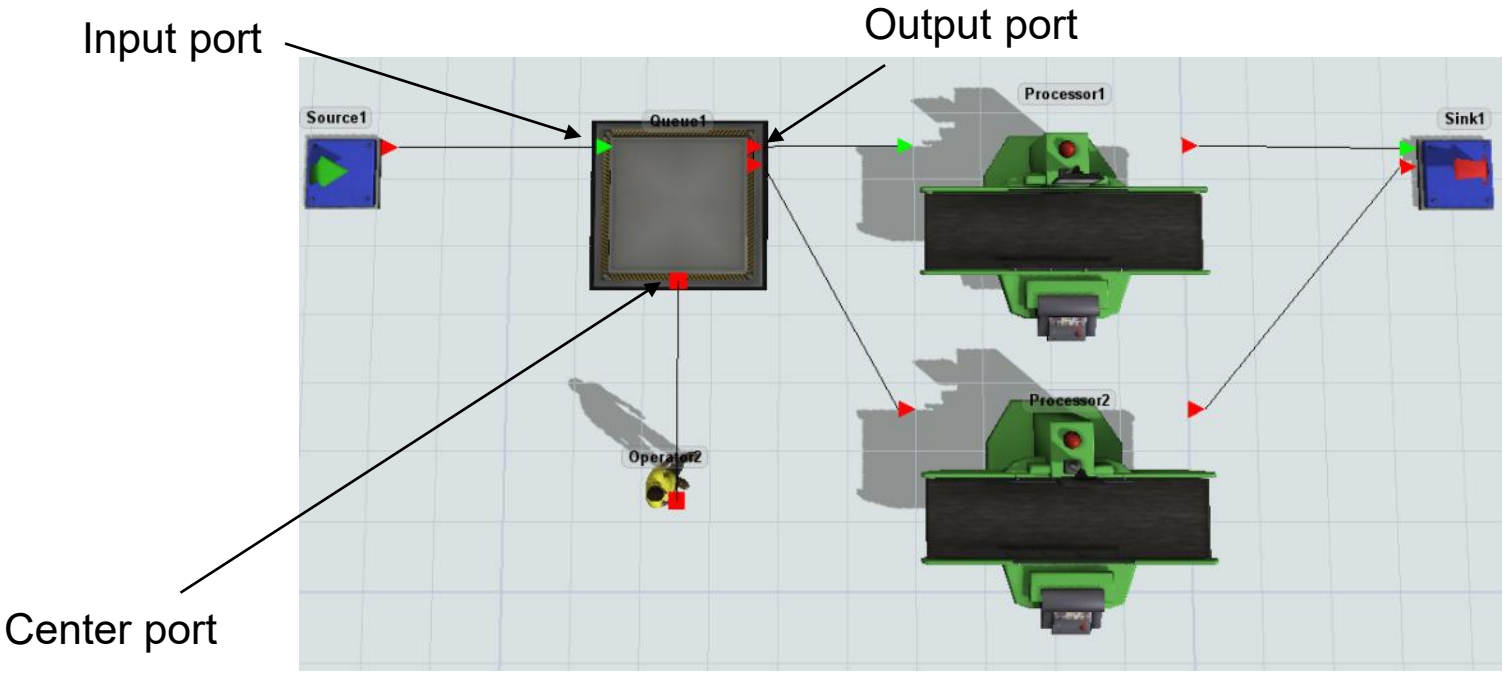
In FlexSim simulation, items move from one Object (Fixed Resource) to another Object.

The direction of flow is from the <u>output port</u> of an object to the <u>input port</u> of the other object.

Port Connections

- **Output/ Input Ports** – Output-to-Input port connection defines direction of flow.

- **Center Ports** – Connection between Task Executer and Fixed Resource. Used to assign Task Executer to perform task or move flowitem.



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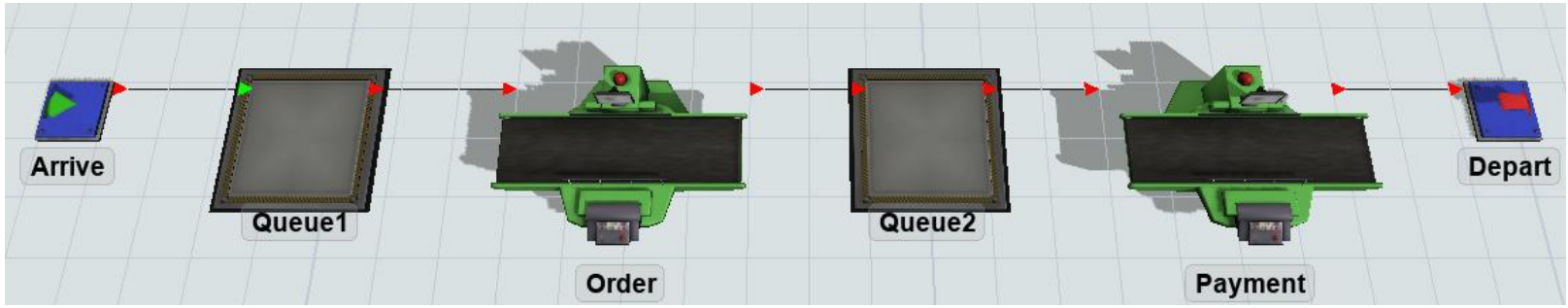
Controls for Port Connections

	Output to Input	Center to Center
Disconnect	Disconnect Objects Q Q	Disconnect Center Ports W W
Connect	Connect Objects A A	Connect Center Ports S S
View	V	C

Build Your 1st FlexSim Model

At a popular food stall, customers go through a two-stage queuing process.

Stage 1: Customer order food - served by staff #1
Stage 2: Payment - served by staff #2

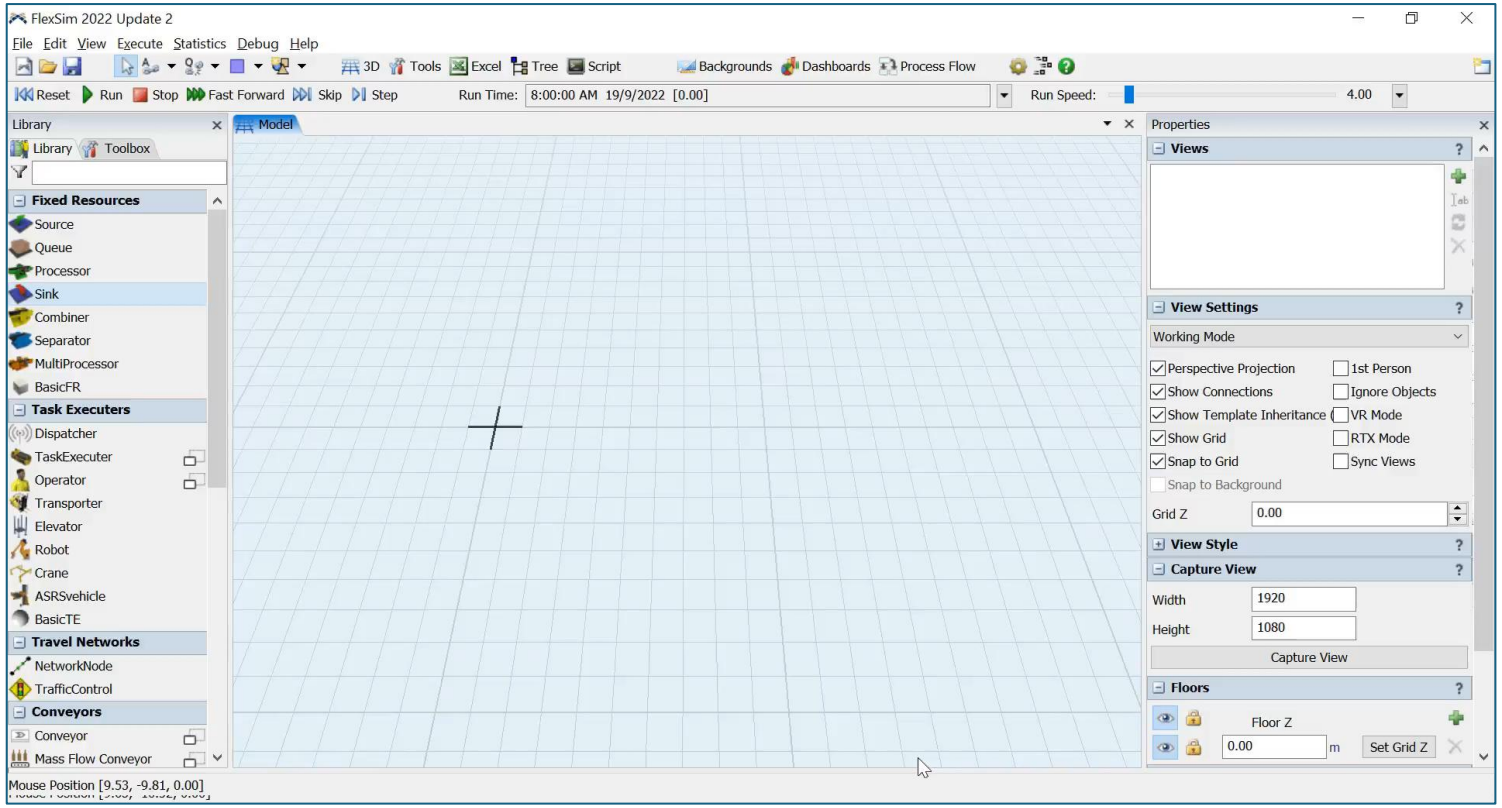


1. Place the objects as shown from left to right

2. Rename the objects

3. Connect the objects accordingly

Connect Objects



Click to see Video

Connect objects:

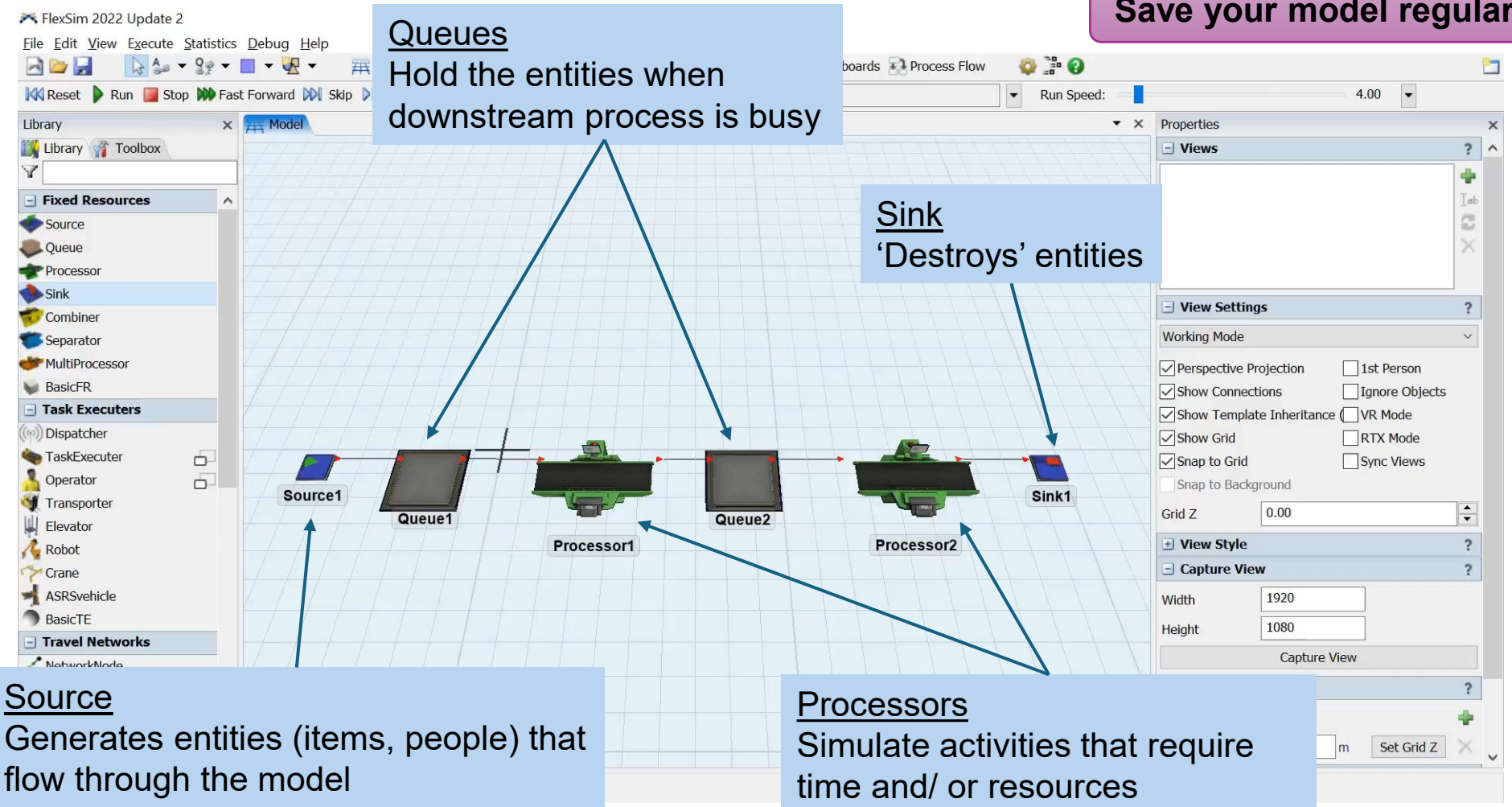
- Use Connect Objects (A) icon in menu bar
- Or Hold down ‘A’ key and click to drag from one object to another in the direction of flow

Disconnect objects:

- Use Disconnect Objects (Q) icon in menu bar
- Or Hold down ‘Q’ key and click drag from one object to another in the direction of connection

Rename Objects

Save your model regularly!



Queues
Hold the entities when downstream process is busy

Sink
'Destroys' entities

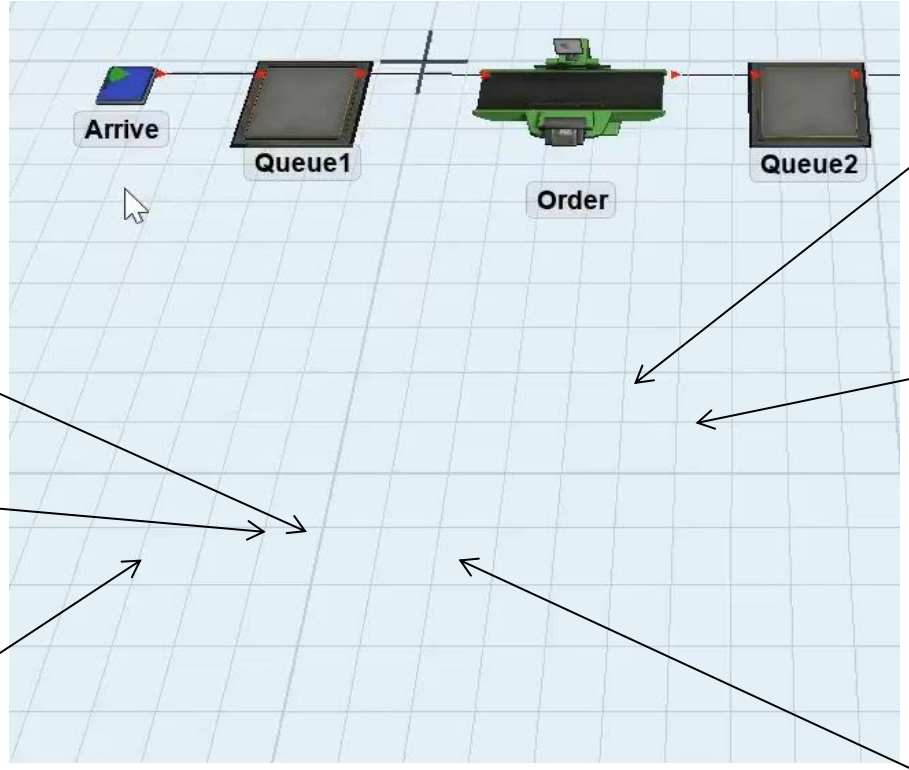
Source
Generates entities (items, people) that flow through the model

Processors
Simulate activities that require time and/ or resources

Click to see Video

Set Source Parameters

On average, 150 people patronize the stall over a period of 3 hrs.
Mean inter-arrival time = 180 mins/ 150 = 1.2 mins



Scale (mean)

Location (min value)

Probability distribution

Random number stream

FlowItem Class:
- Changes the visual (graphics) of FlowItems
- Change to **Man**

Arrival Style:
- **Inter-Arrival Time** – Time between arrivals
- **Arrival Schedule** – Timing & number of arrivals
- **Arrival Sequence** – Sequence of arrivals

Change Inter-arrival Time to **exponential(0, 1.2, getstream(current))**

Click to see Video

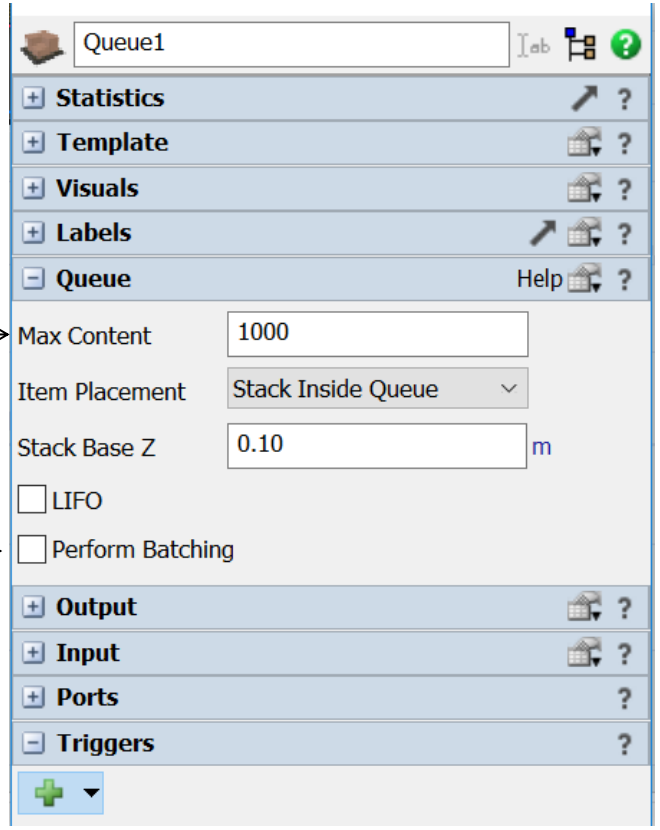
Set Queue Parameters

Queue allows items to be held at a designated space.

This allows items to exit from upstream process even when the downstream process is busy.

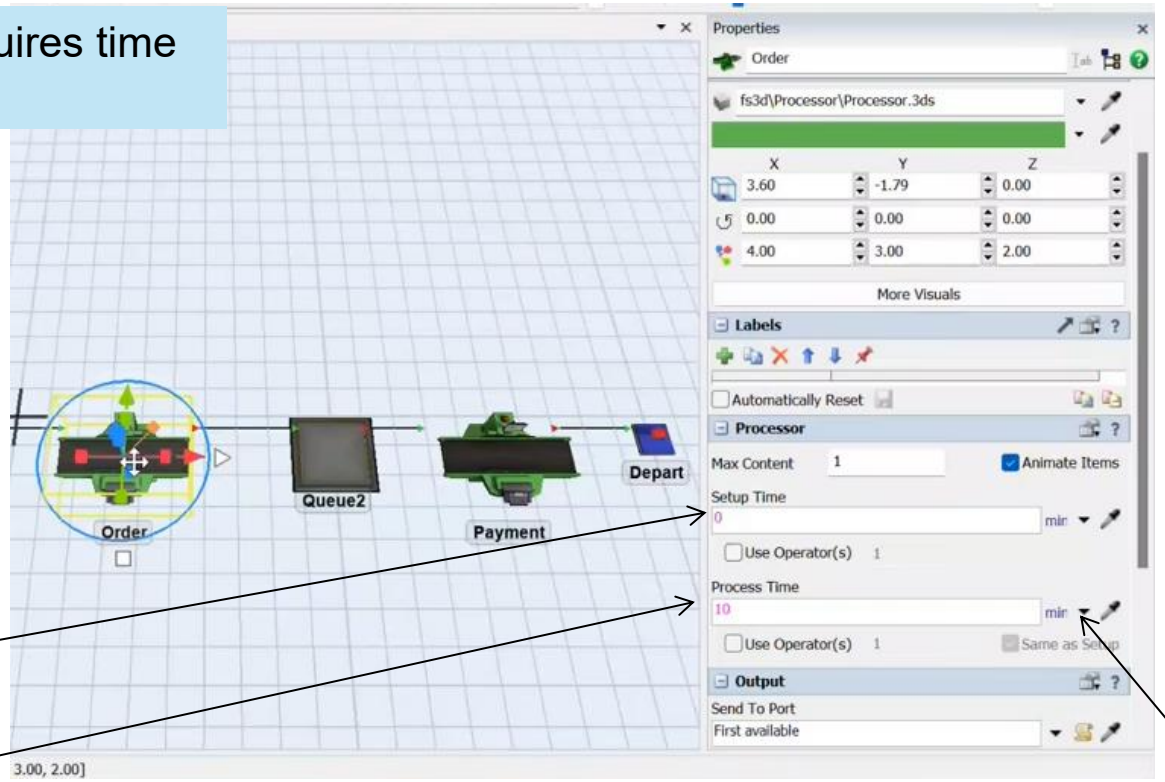
Maximum Content:
- Queue capacity

Perform Batching:
- Batch items before release



Set Processor Parameters

Processor simulates an activity that requires time and/or resource to complete.



Setup Time

Process Time

Click option button to select Statistical Distribution

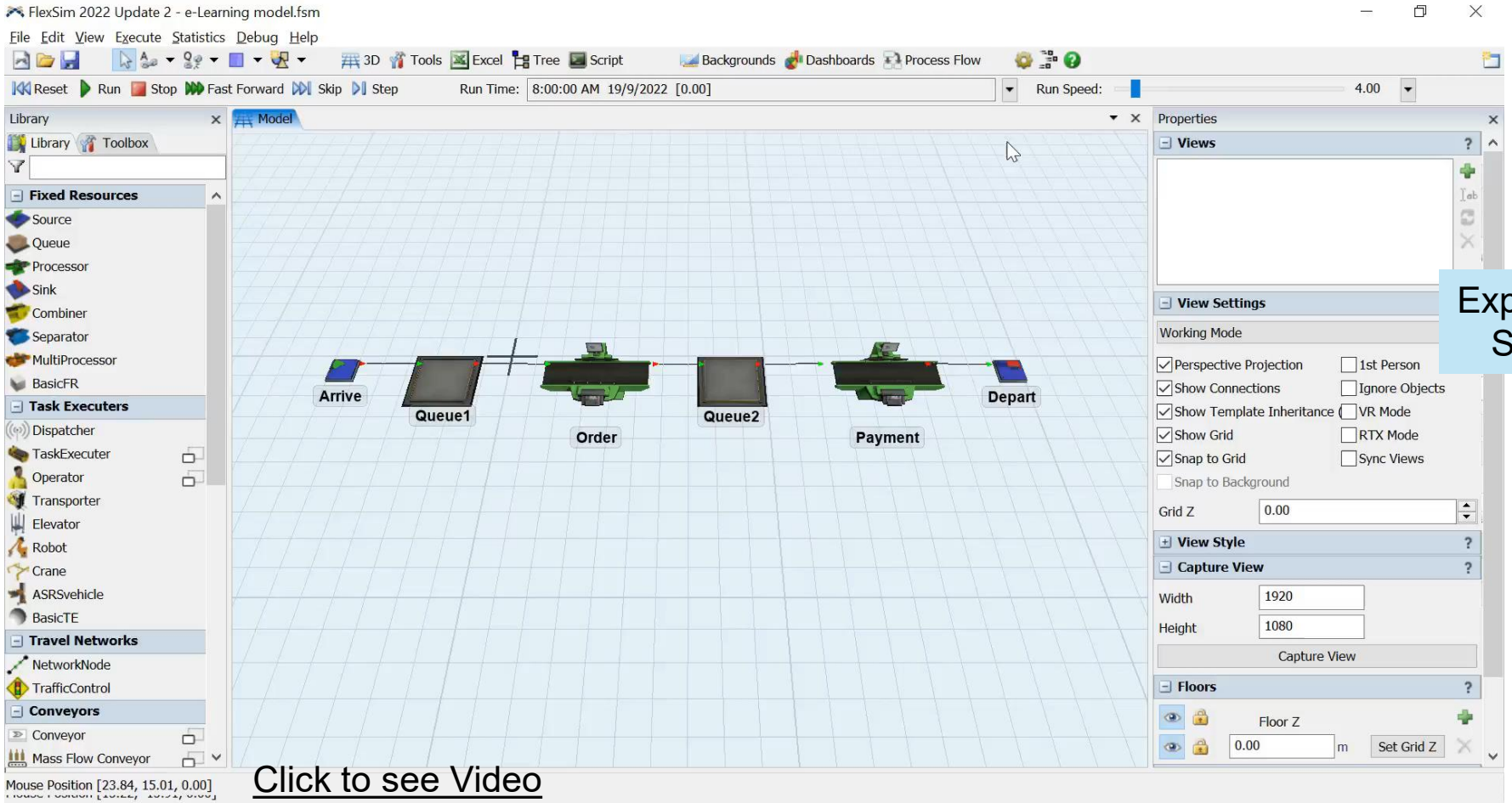
Click to see Video

1. Set Process Time of **Order processor** to **normal(1.5, 0.7, getstream(current))**

2. Set Process Time of **Payment processor** to **triangular(0.5, 2, 1, getstream(current))**

Set Stop Time, Run Speed & Simulate

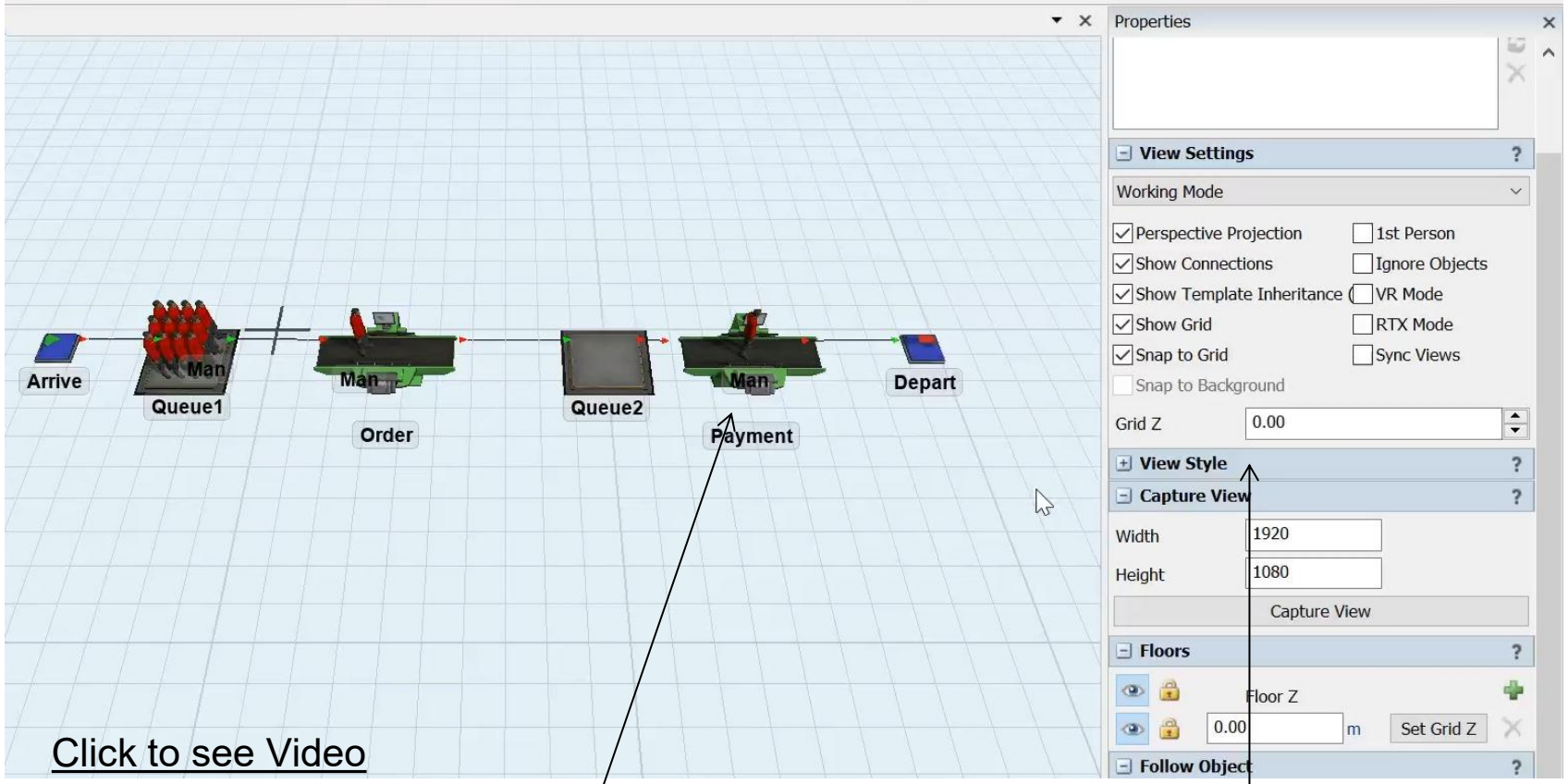
- Set Simulation **Start Time** to **11am** and **Stop Time** to **180 mins** or **2 pm**
- **Run Speed** reflects how fast the simulation is running in multiples of seconds. E.g. Run Speed = 10 will make simulation run 10 seconds for every 1 second in real time.



Click to see Video

Experiment with different Run Speed in your simulation.

View Stats of Objects during Simulation



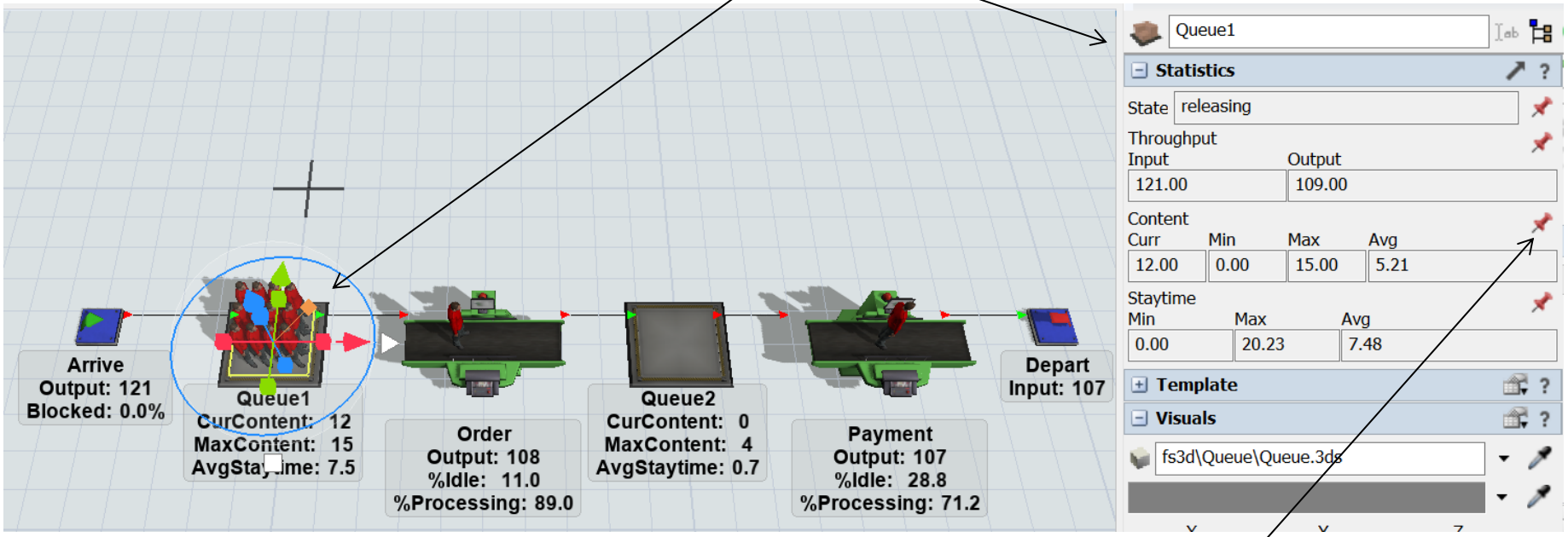
Click to see Video

Space out the objects to ‘unblock’ stats

Under View Style, select show names & stats

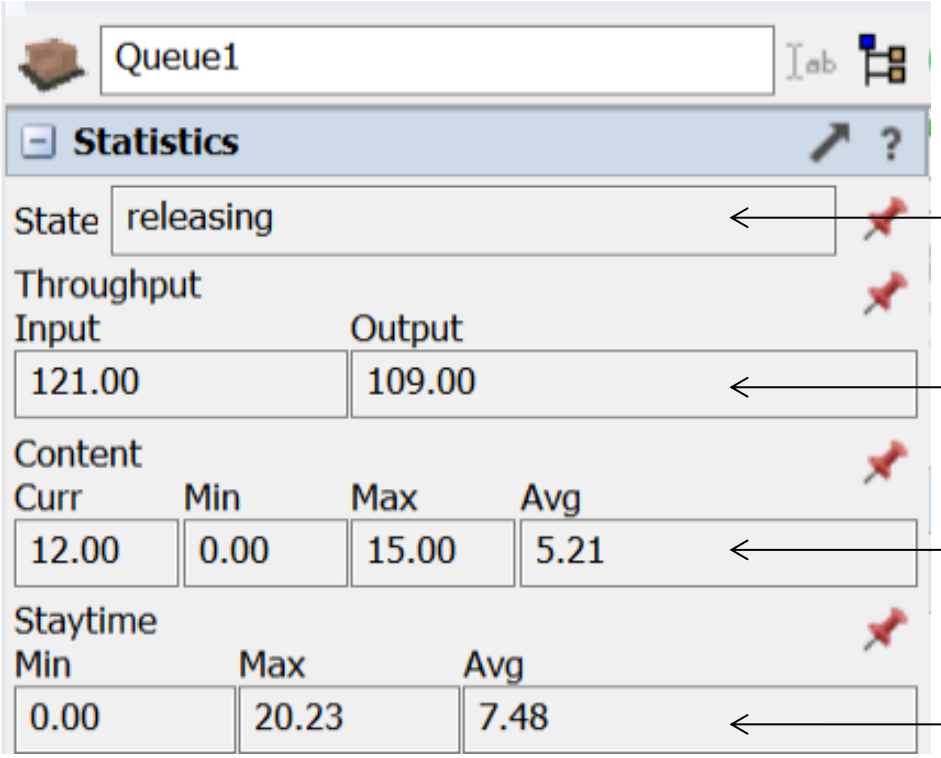
View Stats in Quick Properties Window

Click on a Fixed Resource to view its object statistics in Properties window



Click here to pop out Statistics page

Interpreting Object Statistics



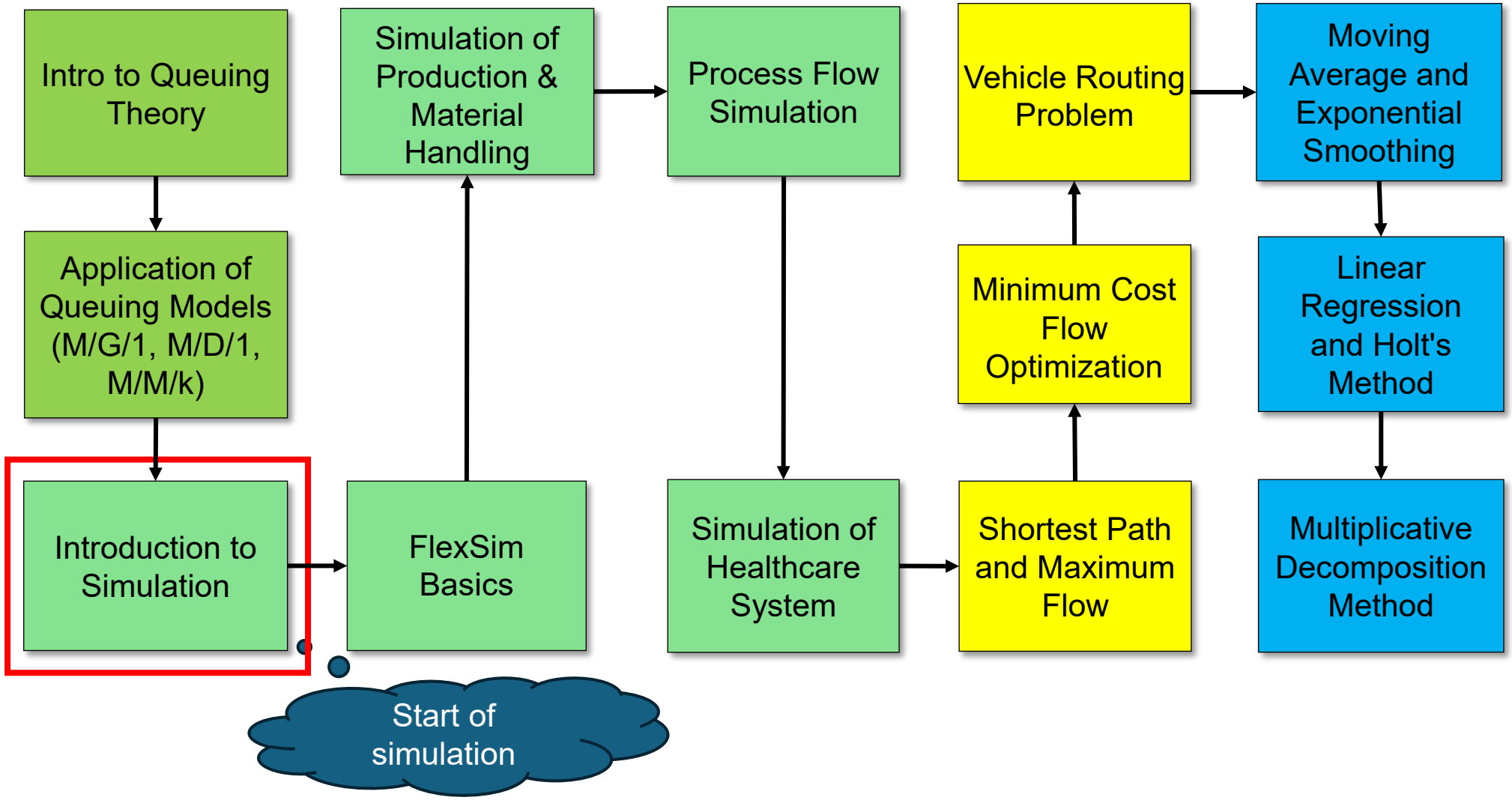
State: What the object is doing

Throughput: How many items have entered (Input) and exit (Output)

Content: How many items are there inside the object? (Current, Minimum, Maximum & Average)

Staytime: How long did each item stay in the object? (Minimum, Maximum & Average)

E211 Progression



SCORM Quiz

Quiz

Click the **Quiz** button to edit this object

Identify the type of simulation seen in each video from SCORM page 7.

Video 1

Video 2

Video 3

Video 4

Continuous simulation | Discrete event simulation | Continuous simulation

Monte Carlo simulation